

Assessing pre-training bias in Health data and estimating its impact on machine learning algorithms

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Abstract. *This meta-paper describes the style to be used in articles and short papers for SBC conferences. For papers in English, you should add just an abstract while for the papers in Portuguese, we also ask for an abstract in Portuguese (“resumo”). In both cases, abstracts should not have more than 10 lines and must be in the first page of the paper.*

1. Introduction

Machine learning (ML) is increasingly used in various domains, including Health, where it assists in diagnosis, prognosis, and treatment [Chen et al. 2021]. However, ML algorithms can potentially introduce biases, leading to unfair and potentially harmful outcomes [Obermeyer et al. 2019, Juhn et al. 2022].

This work aims to measure the risk of bias in Health datasets using pre-training bias metrics and assesses their impact on the performance of supervised ML algorithms, correlating the bias metrics to the overall models’ performance. We analyze four pre-training bias metrics [Hardt et al. 2021]: Class Imbalance (CI), Kullback-Leibler (KL) Divergence, Kolmogorov-Smirnov (KS), and Conditional Demographic Disparity in Labels (CDDL).

1.1. Class Imbalance (CI)

CI measures the representation of different classes within a dataset. It ranges from -1 to 1, with values near zero indicating balanced representation. Positive values indicate overrepresentation of the advantaged group, while negative values indicate overrepresentation of the disadvantaged group.

1.2. Kullback-Leibler (KL) Divergence

KL Divergence measures the divergence between the label distributions of two facets. It ranges from 0 to $+\infty$, with values near zero indicating similar distributions and higher values indicating greater divergence.

1.3. Kolmogorov-Smirnov (KS)

KS measures the maximum divergence between labels in the distribution for different facets. It ranges from 0 to 1, with values near zero indicating evenly distributed labels and values near one indicating unbalanced labels.

1.4. Conditional Demographic Disparity in Labels (CDDL)

CDDL measures the proportion of negative outcomes in a specific facet of a dataset. It ranges from -1 to 1, with positive values indicating demographic disparity favoring the advantaged group and negative values indicating disparity favoring the disadvantaged group.

2. References

References

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